

How Much EEG Data Is Enough?

Finding the Sweet Spot for Stable Auditory Neural Tracking

Jaewon Lee^{1, 2}, Jonghwa Jeonglok Park³, Yoonseob Lim³

1. Department of Composition, Seoul National University, Seoul, South Korea.

2. Department of Computer Science and Engineering, Seoul National University, Seoul, South Korea.

3. Intelligence and Interaction Research Center, Korea Institute of Science and Technology, Seoul, South Korea.



1. Introduction

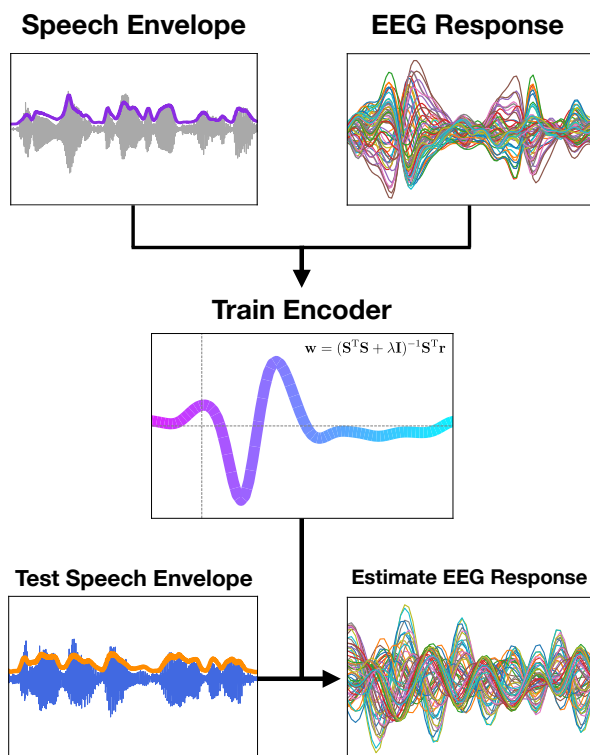
- The **Temporal Response Function (TRF)**, an **encoder model** linking sensory stimulus features (**speech envelope**) to neural responses (**EEG**), provides insights into how the brain processes complex stimulus properties.
- How much EEG data** do we need to ensure **reliable encoding performance**?

2. Limitations

- Constructing an **encoder** with reliable performance and stability depends on the **quantity of data used for training**.
- Practical challenges** such as participant fatigue, discomfort in EEG acquisition, and excessive noise or artifacts in EEG often **limit the amount of usable data**, resulting in an **unstable encoder**.
- We investigate the **impact of data quantity on encoder stability** using two EEG datasets from speech-listening tasks.

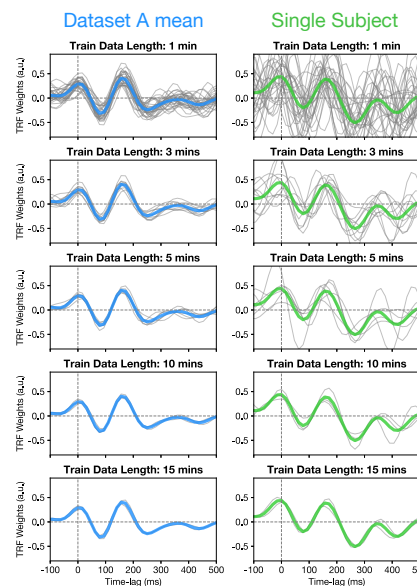
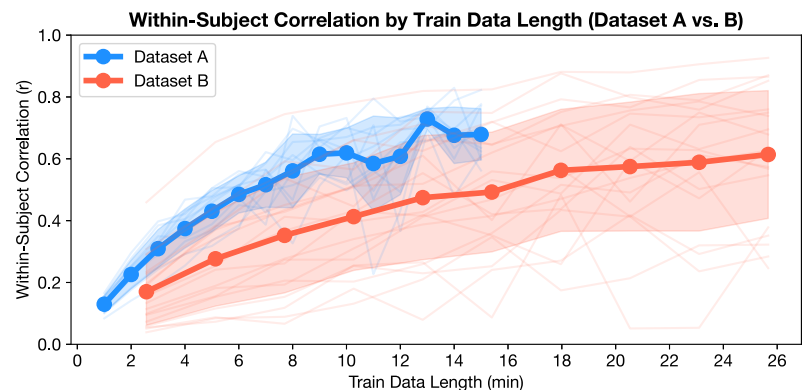
3. Methods

- Two publicly available datasets were used in this study.
 - Dataset A (KIST)**⁵
 - Dichotic Speech** (Cocktail Party) Stimuli, **Korean**
 - 30 subjects × 30 minutes
 - Dataset B (Lalor Lab)**³
 - Single Speech** Stimuli, **English**
 - 19 subjects × 51 minutes
 - The **encoder** is modeled as follows:
- $$\text{EEG} = \text{Encoder} * \text{Speech Envelope} + \varepsilon \text{ (residuals)}$$



4. Procedure & Results

- We used monotonically increasing train data durations to train the **encoders**, ensuring no duplicate trials within the same duration unit.
- An additional **encoder**, trained with the maximum available data, served as **Ground Truth**.



- We compute the **average of the pairwise correlations of encoders** within each subject for each duration unit. These **within-subject correlations** are then **averaged** across all participants.

Our findings reveal a **highly positive relationship** between data quantity and **within-subject encoder correlation**:

Dataset A: $r = 0.943$, $p < 0.001$,
Dataset B: $r = 0.965$, $p < 0.001$.

References

- Bialas, O., Dou, J., & Lalor, E. C. (2023). mTRFpy: A Python package for temporal response function analysis. Journal of Open Source Software, 8(89), 5657.
- Crosse, M. J., Di Liberto, G. M., Bednar, A., & Lalor, E. C. (2016). The multivariate temporal response function (mTRF) toolbox: a MATLAB toolbox for relating neural signals to continuous stimuli. Frontiers in Human Neuroscience, 10, 604.
- Di Liberto, G. M., O'Sullivan, J. A., & Lalor, E. C. (2015). Low-frequency cortical entrainment to speech reflects phoneme-level processing. Current Biology, 25(19), 2457-2465.
- Mesik, J., & Wojtczak, M. (2023). The effects of data quantity on performance of temporal response function analyses of natural speech processing. Frontiers in Neuroscience, 16, 963629.
- Park, J. J., Baek, S. C., Suh, M. W., Choi, J., Kim, S. J., & Lim, Y. (2023). The effect of topic familiarity and volatility of auditory scene on selective auditory attention. Hearing Research, 433, 108770.

Conclusion

- Higher data quantity resulted in lower weight variance, **reinforcing the stability of the encoders**.
- Our findings emphasize the importance of **optimizing data quantity to balance model stability and generalization**, ensuring reliable **encoder** performance across different subjects and conditions.